MATH O MAGIC

1·2

by

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and
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INTRODUCTION

MATH O MAGIC 1·2 is part of a series of 12 books

MATHOMAGIC 1·2	MATHOMAGIC 4·3
MATHOMAGIC 1·3	MATHOMAGIC 4·4
MATHOMAGIC 2·3	MATHOMAGIC 5·3
MATHOMAGIC 2·4	MATHOMAGIC 5·4
MATHOMAGIC 3·3	MATHOMAGIC 6·3
MATHOMAGIC 3·4	MATHOMAGIC 6·4

These books are designed for students in years eight to ten.

1·2 and 1·3 are designed to revise the skills from earlier years and to be a platform from which high school skills are developed.

2·3, 3·3, 4·3, 5·3 and 6·3 develop skills in the SPACE, MEASUREMENT, and INFERENCE strands.

2·4, 3·4, 4·4, 5·4 and 6·4 address skill in the NUMBER and FUNCTION strands.

Stages 2, 3, 4, 5 and 6 indicate the level of difficulty, with each following in a sequential manner.

PREFACE

Often students find the necessary skill practice in Mathematics to be very boring. This series of books has been designed as a motivational tool to help the students to feel more enthusiastic about their work.

These worksheets are designed to be used in various ways –

1. MOTIVATION – Students are keen to solve the puzzle and keep asking for more.

2. INCENTIVE – As a reward for students who finish their work early.

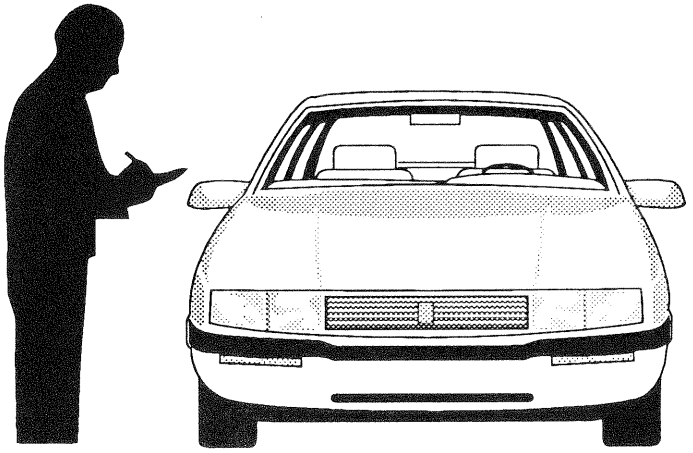
3. SELF CHECKING
   – For students to be able to check their own work as they go along.

4. EXTRA PRACTICE
   – A fun way of doing extra practice on skills development.

5. HOMEWORK – an interesting way to get students to more consolidation of mathematical skills in a way they enjoy.

6. REVISION – An exciting way to stimulate revision.

However you use these sheets, you will always be asked for more. The students enjoy working out the puzzles and riddles and get a great sense of satisfaction from getting them right.



To answer this riddle, find the best approximation for each question in the boxes below. Write the letter of the question above the approximation.

WHAT DID THE PARKING INSPECTOR SAY AFTER BOOKING ONE HUNDRED MOTORISTS FOR ILLEGAL PARKING?

K 143 ÷ 7 is about . . .

E 31 x 8 is about . . .

E 674 + 439 is about . . .

S 461 – 194 is about . . .

V 443 ÷ 20 is about . . .

A 18 x 7 is about . . .

I 976 + 440 is about . . .

Y 946 – 250 is about . . .

I 636 ÷ 21 is about . . .

W 78 x 7 is about . . .

D 998 + 508 is about . . .

N 674 – 275 is about . . .

O 494 ÷ 50 is about . . .

D 26 x 40 is about

R 75 + 122 is about . . .

A 313 – 160 is about . . .

N 989 ÷ 21 is about . . .

E 63 x 29 is about . . .

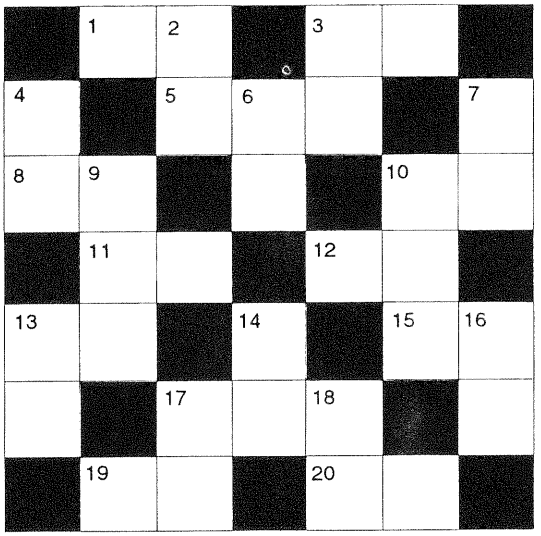
O 477 + 331 is about . . .

F 955 – 347 is about . . .

1400	22	250		1500	800	50	1100		130		600	30	400	1800

1000	150	700	270		550	10	200	20

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MULTIPLICATION OF DECIMALS

WRITE THE DECIMAL POINT ON THE SAME SQUARE AS THE FIRST DECIMAL PLACE

ACROSS

1. 2·5 x 3·64

3. 78 x 0·5

5. 2·2 x 3·3

8. 2·4 x 3·5

10. 5 x 10·2

11. 8 x 1·2

12. 1·5 x 3·2

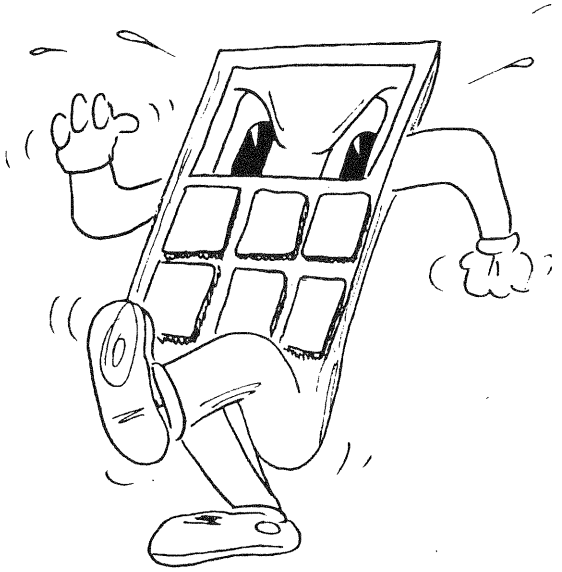
13. 0·074 x 1000

15. 44 x 1·25

17. 1·1 x 1·1

19. 4 x 0·4

20. 0·4 x 2·2



DOWN

2. 6·8 x 0·025

3. 5 x 7·2

4. 880 x 0·1

6. $0{\cdot}5^2$

7. 90 x 0·9

9. 2·47 x 0·2

10. 4 x 1·4625

13. 8 x 0·95

14. 4 x 2·3

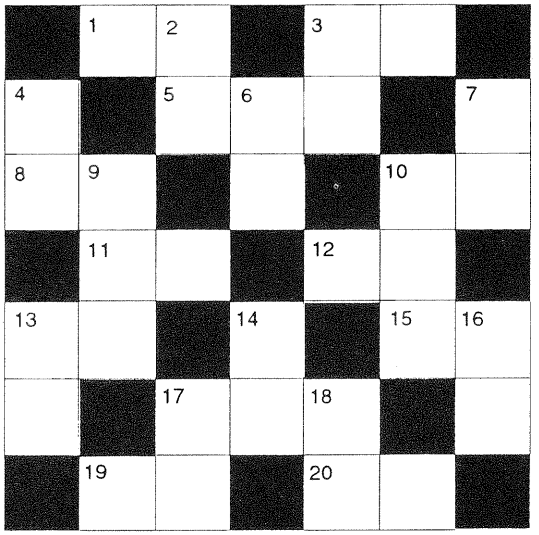
16. 7 x 0·8

17. 8 x 0·2

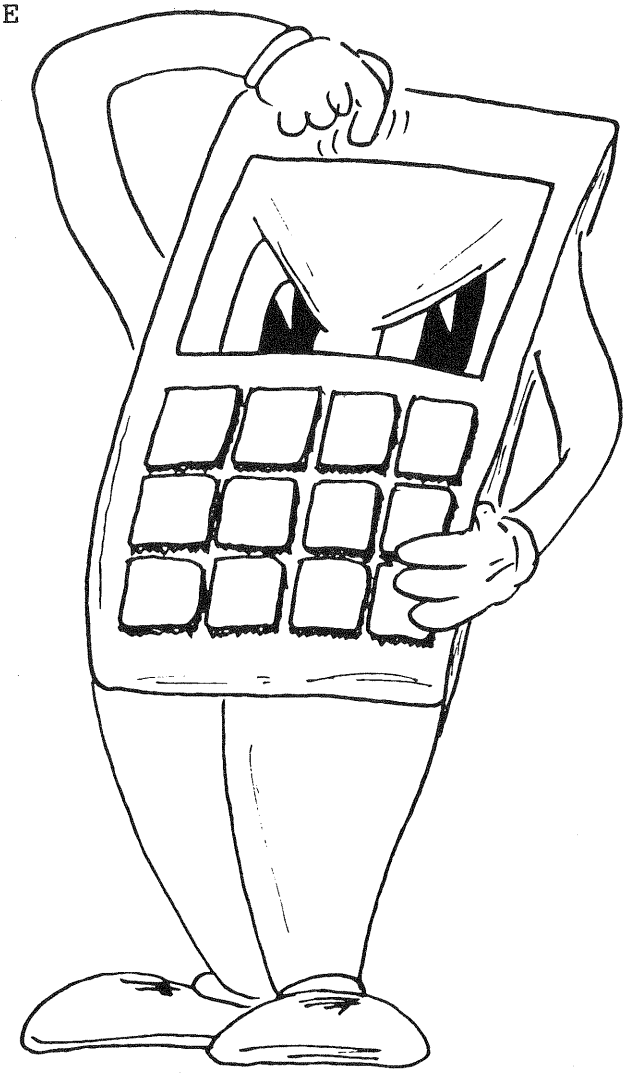
18. 2·25 x 0·8

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DIVISION of DECIMALS



WRITE THE DECIMAL POINT ON THE SAME SQUARE AS THE FIRST DECIMAL PLACE



ACROSS

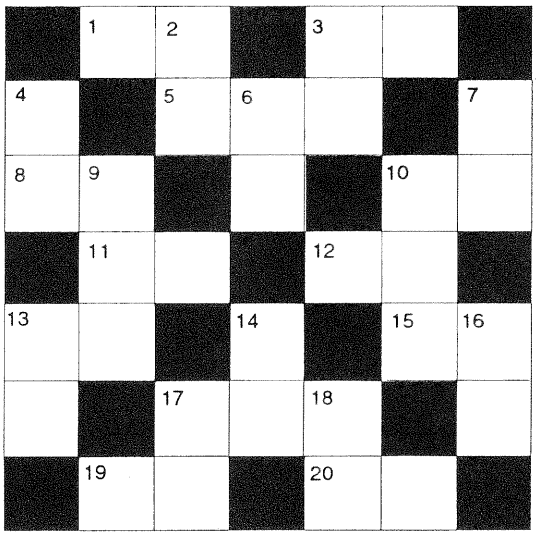
1. 44 ÷ 10
3. 88 ÷ 80
5. 204·8 ÷ 4
8. 310 ÷ 5
10. 372 ÷ 6
11. 0·75 ÷ 5
12. 192 ÷ 8
13. 2 ÷ 8
15. 1·17 ÷ 9
17. 196 ÷ 80
19. 18 ÷ 4
20. 3·54 ÷ 6

DOWN

2. 0·9 ÷ 2
3. 4·8 ÷ 4
4. 8600 ÷ 100
6. 6 ÷ 4
7. 13·2 ÷ 0·6
9. 12·9 ÷ 6
10. 192·3 ÷ 3
13. 220 ÷ 1000
14. 13·6 ÷ 4
16. 136 ÷ 4
17. 10 ÷ 4
18. 33 ÷ 6

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MULTIPLICATION AND DIVISION OF DECIMALS



ACROSS

1. 9 x 1·1

3. 0·7 x 0·7

5. 26·25 ÷ 5

8. 1·15 x 6

10. 20 ÷ 8

11. 15·2 ÷ 8

12. 33 x 0·3

13. 9·9 ÷ 0·9

15. 6 x 0·9

17. 49·5 ÷ 6

19. 99 ÷ 90

20. 69 ÷ 10

DOWN

2. 7·6 ÷ 8

3. 9 ÷ 20

4. 2·4 x 40

6. 0·72 ÷ 3

7. 9 ÷ 6

9. 911 ÷ 1000

10. 0·295 x 10

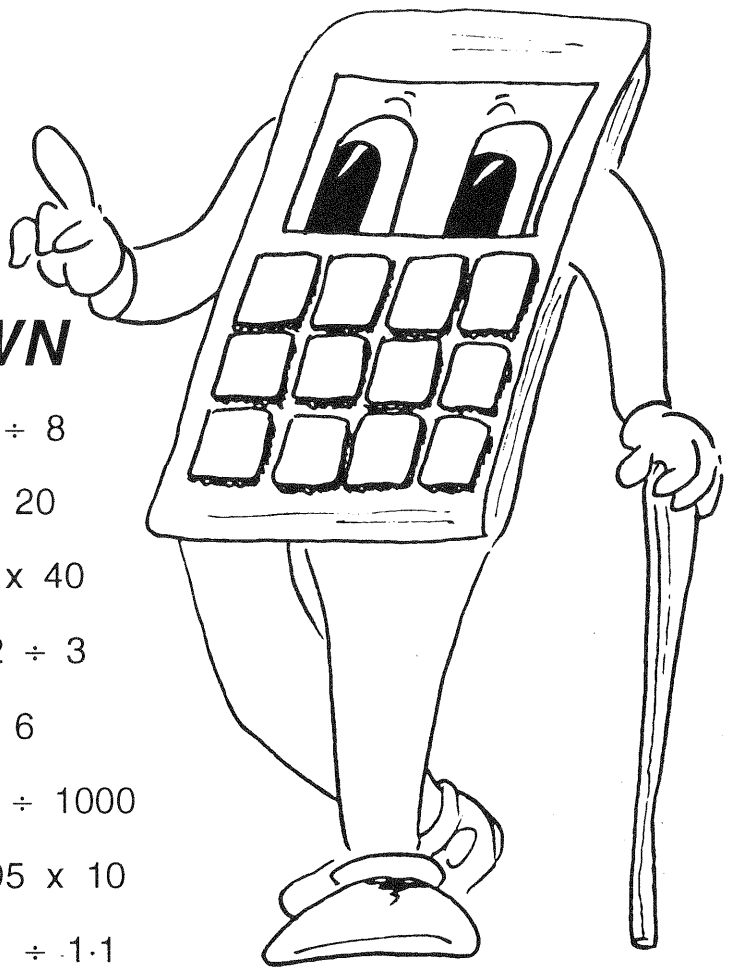
13. 1·21 ÷ 1·1

14. 35·2 ÷ 11

16. 9 x 0·05

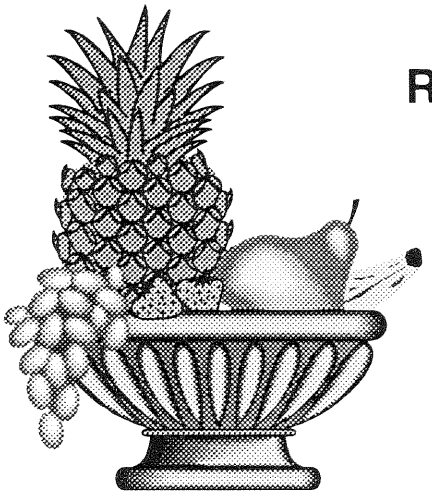
17. 0·9 x 9

18. 0·056 x 1000



WRITE THE DECIMAL POINT ON THE SAME SQUARE AS THE FIRST DECIMAL PLACE

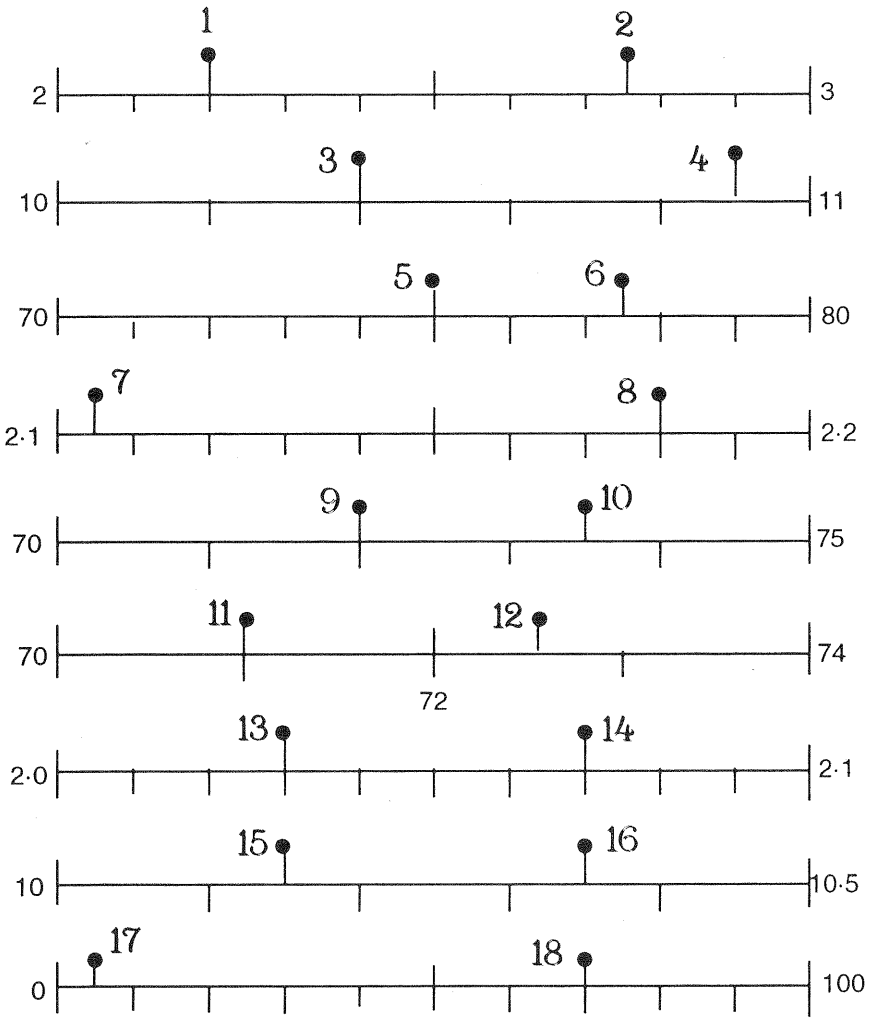
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READING SCALES

Find the values indicated by the DOTS on each of the scales and look for that value in the code box. Note the letter next to the value in the code box. Place this letter above the question number in the answer box to solve the riddle:

WHY DON'T GRAPE GROWERS GO TO THE DOCTOR?



CODE	
E	2·105
E	73·5
E	5
E	2·03
A	2·75
A	2·07
I	10·15
L	2·18
L	10·4
F	75
T	72
W	2·2
H	10·9
V	70
N	71
Y	77·5
Y	10·35
S	72·5

9	4	13	6		2	8	1	14	16	12

5	10	17	3		18	15	11	7

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Perform the following conversions and match the question number to the answer in the code box. Use this information to find the answer to the question:

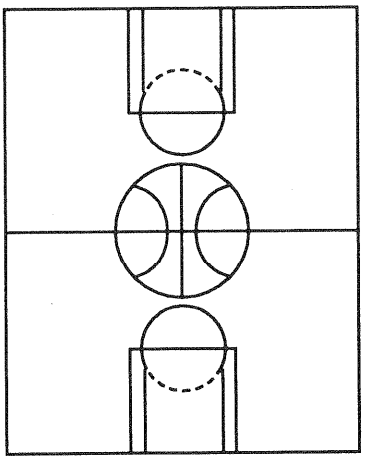
WHAT DISEASE DO YOU GET FROM PLAYING BASKETBALL?

CODE BOX	
O	1 000
O	100 000
O	100
P	0·001
I	0·01
U	1
H	0·000 001
H	0·0001
C	0·000 01
G	1 000 000
G	10 000
N	10

1. Convert 1 m to centimetres.
2. Convert 1 m to millimetres.
3. Convert 1 m to kilometres.
4. Convert 1 cm to metres.
5. Convert 1 cm to millimetres.
6. Convert 1 cm to kilometres.
7. Convert 1 km to centimetres.
8. Convert 1 km to millimetres.
9. Convert 10 km to metres.
10. Convert 10 mm to centimetres.
11. Convert 0·1 mm to metres.
12. Convert 1 mm to kilometres.

11	1	2	3	4	5	9

6	7	10	8	12



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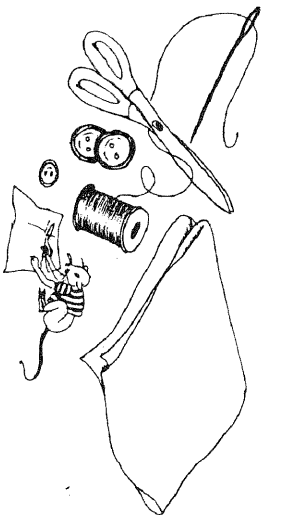
WHAT DID THE GENIE SAY AT THE LAUNDROMAT?

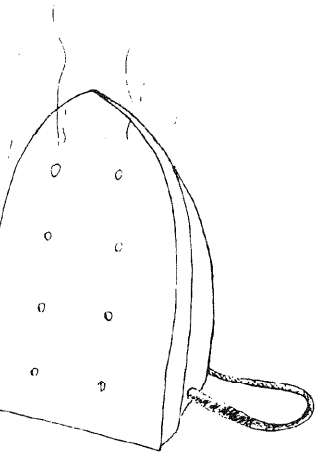


To solve the puzzle, find the pairs of equivalent fractions and note the number and letter that go together. Write the letter above the number to reveal the answer.



1	$\frac{1}{2}$	•	•	$\frac{3}{9}$ E
2	$\frac{3}{4}$	•	•	$\frac{70}{100}$ Y
3	$\frac{2}{3}$	•	•	$\frac{42}{49}$ I
4	$\frac{1}{4}$	•	•	$\frac{24}{27}$ L
5	$\frac{5}{6}$	•	•	$\frac{2}{10}$ S
6	$\frac{9}{10}$	•	•	$\frac{9}{15}$ U
7	$\frac{3}{8}$	•	•	$\frac{49}{56}$ R
8	$\frac{1}{3}$	•	•	$\frac{5}{20}$ O
9	$\frac{5}{8}$	•	•	$\frac{10}{12}$ L
10	$\frac{4}{9}$	•	•	$\frac{6}{9}$ T
11	$\frac{1}{6}$	•	•	$\frac{5}{10}$ A
12	$\frac{8}{9}$	•	•	$\frac{9}{30}$ A
13	$\frac{6}{7}$	•	•	$\frac{30}{40}$ E
14	$\frac{7}{10}$	•	•	$\frac{27}{30}$ H
15	$\frac{3}{5}$	•	•	$\frac{6}{27}$ W
16	$\frac{5}{6}$	•	•	$\frac{10}{16}$ G
17	$\frac{1}{5}$	•	•	$\frac{20}{25}$ R
18	$\frac{7}{8}$	•	•	$\frac{4}{24}$ S
19	$\frac{2}{9}$	•	•	$\frac{15}{40}$ T
20	$\frac{1}{8}$	•	•	$\frac{16}{36}$ N
21	$\frac{3}{10}$	•	•	$\frac{15}{18}$ H
22	$\frac{4}{5}$	•	•	$\frac{3}{24}$ E

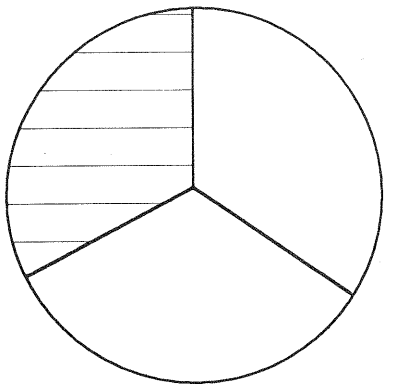




13, 5, 12, 9, 22, 1, 10, 3 14, 4, 15, 7, 16, 18, 2, 8, 19, 21, 11, 6, 20, 17.

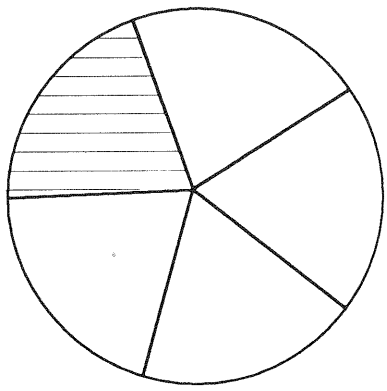
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Look for equivalent fractions to the one shown and shade in the boxes containing them. If you find them all a message will appear.



$\frac{1}{3}$

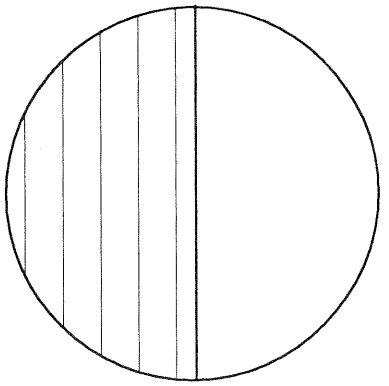
$\frac{2}{6}$	$\frac{6}{18}$	$\frac{13}{39}$	$\frac{15}{45}$	$\frac{3}{9}$
$\frac{14}{42}$	$\frac{5}{9}$	$\frac{5}{8}$	$\frac{2}{5}$	$\frac{12}{36}$
$\frac{8}{24}$	$\frac{3}{5}$	$\frac{8}{9}$	$\frac{2}{3}$	$\frac{7}{21}$
$\frac{11}{33}$	$\frac{9}{10}$	$\frac{7}{8}$	$\frac{3}{4}$	$\frac{9}{27}$
$\frac{10}{30}$	$\frac{3}{9}$	$\frac{1}{3}$	$\frac{20}{60}$	$\frac{5}{15}$



$\frac{1}{5}$

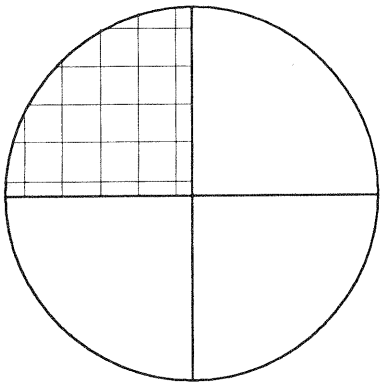
$\frac{2}{10}$	$\frac{3}{4}$	$\frac{4}{9}$	$\frac{5}{9}$	$\frac{5}{25}$
$\frac{8}{40}$	$\frac{1}{10}$	$\frac{5}{10}$	$\frac{15}{20}$	$\frac{9}{45}$
$\frac{6}{30}$	$\frac{13}{65}$	$\frac{11}{55}$	$\frac{15}{75}$	$\frac{7}{35}$
$\frac{10}{50}$	$\frac{5}{8}$	$\frac{1}{3}$	$\frac{1}{2}$	$\frac{12}{60}$
$\frac{4}{20}$	$\frac{3}{8}$	$\frac{1}{8}$	$\frac{4}{5}$	$\frac{3}{15}$

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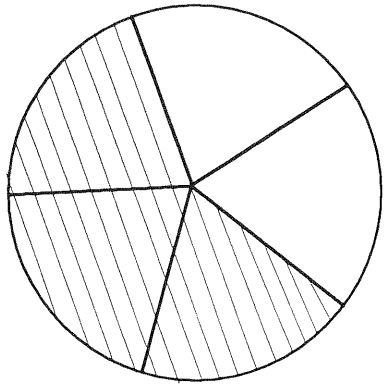
$\frac{1}{2}$

$\frac{2}{4}$	$\frac{12}{24}$	$\frac{6}{12}$	$\frac{15}{30}$	$\frac{8}{16}$
$\frac{9}{18}$	$\frac{5}{6}$	$\frac{1}{3}$	$\frac{11}{22}$	$\frac{3}{5}$
$\frac{4}{9}$	$\frac{3}{4}$	$\frac{4}{8}$	$\frac{1}{5}$	$\frac{5}{8}$
$\frac{6}{10}$	$\frac{20}{40}$	$\frac{8}{9}$	$\frac{7}{8}$	$\frac{3}{10}$
$\frac{3}{6}$	$\frac{10}{20}$	$\frac{7}{14}$	$\frac{13}{26}$	$\frac{5}{10}$



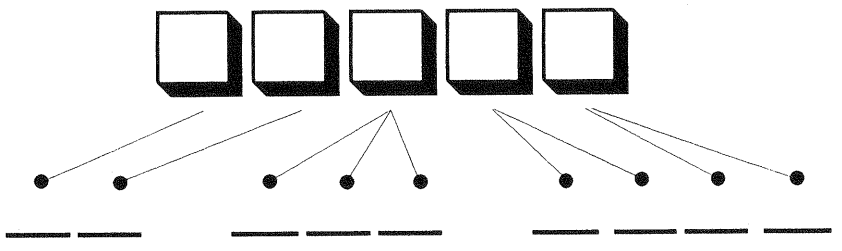
$\frac{1}{4}$

$\frac{6}{24}$	$\frac{25}{100}$	$\frac{9}{36}$	$\frac{15}{60}$	$\frac{3}{12}$
$\frac{12}{48}$	$\frac{1}{6}$	$\frac{1}{16}$	$\frac{2}{9}$	$\frac{3}{4}$
$\frac{8}{32}$	$\frac{10}{40}$	$\frac{4}{16}$	$\frac{1}{5}$	$\frac{4}{9}$
$\frac{13}{52}$	$\frac{5}{6}$	$\frac{3}{8}$	$\frac{1}{2}$	$\frac{3}{16}$
$\frac{5}{20}$	$\frac{11}{44}$	$\frac{7}{28}$	$\frac{20}{80}$	$\frac{2}{8}$



$\frac{3}{5}$

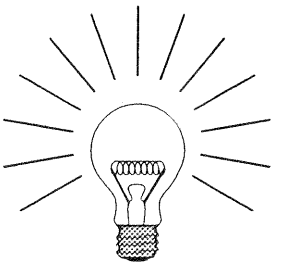
$\frac{21}{35}$	$\frac{33}{55}$	$\frac{15}{25}$	$\frac{24}{40}$	$\frac{6}{10}$
$\frac{30}{50}$	$\frac{3}{8}$	$\frac{2}{3}$	$\frac{15}{16}$	$\frac{13}{15}$
$\frac{45}{75}$	$\frac{9}{12}$	$\frac{4}{7}$	$\frac{2}{5}$	$\frac{5}{6}$
$\frac{18}{30}$	$\frac{3}{15}$	$\frac{1}{2}$	$\frac{7}{8}$	$\frac{2}{9}$
$\frac{9}{15}$	$\frac{27}{45}$	$\frac{36}{60}$	$\frac{60}{100}$	$\frac{12}{20}$



Sound out the last three boxes to find the expanded answer in words.

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WHAT HAPPENED TO THE DISCOVERER OF ELECTRICITY?



To answer this question, solve each problem below. Write the letter of each question above the answer. All answers need to be simplified.

A $\frac{3}{5} - \frac{1}{5}$

G $\frac{7}{8} - \frac{3}{8}$

S $\frac{9}{10} - \frac{7}{10}$

A $\frac{11}{12} - \frac{2}{12}$

T $\frac{9}{10} - \frac{1}{10}$

C $\frac{11}{14} - \frac{1}{14}$

H $\frac{11}{12} - \frac{1}{12}$

N $\frac{7}{12} - \frac{5}{12}$

Y $\frac{5}{9} - \frac{4}{9}$

H $\frac{5}{7} - \frac{3}{7}$

E $\frac{7}{8} - \frac{5}{8}$

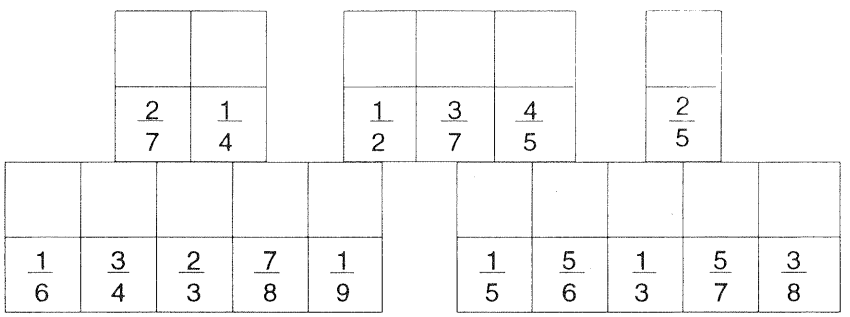
O $\frac{11}{12} - \frac{7}{12}$

S $\frac{8}{9} - \frac{2}{9}$

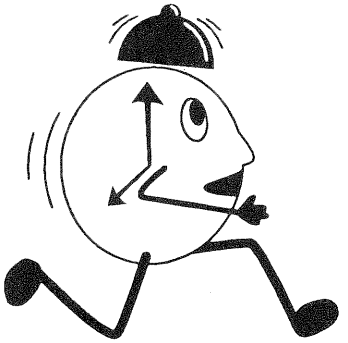
O $\frac{6}{7} - \frac{3}{7}$

T $\frac{15}{16} - \frac{1}{16}$

K $\frac{15}{16} - \frac{9}{16}$



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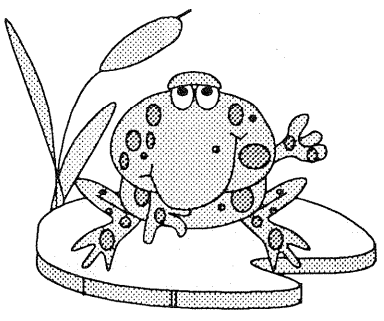


What Is Green And Jumps Two Metres Every Five Seconds?

Add the fractions and then look for your answer in the code below. Wherever you see the answer write the letter of the question above it. All answers have been simplified.

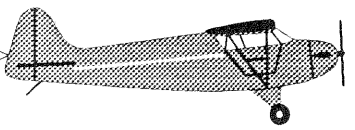
C	$\frac{1}{5} + \frac{3}{5}$	**A**	$\frac{3}{7} + \frac{2}{7}$
G	$\frac{4}{9} + \frac{3}{9}$	**T**	$\frac{2}{5} + \frac{1}{5}$
R	$\frac{1}{10} + \frac{3}{10}$	**P**	$\frac{1}{6} + \frac{3}{6}$
W	$\frac{2}{5} + \frac{3}{5}$	**U**	$\frac{4}{9} + \frac{6}{9}$
I	$\frac{3}{8} + \frac{7}{8}$	**H**	$\frac{3}{10} + \frac{9}{10}$
S	$\frac{1}{4} + \frac{1}{4}$	**C**	$\frac{3}{4} + \frac{3}{4}$
I	$\frac{6}{9} + \frac{6}{9}$	**F**	$\frac{5}{9} + \frac{6}{9} + \frac{7}{9}$
H	$\frac{1}{6} + \frac{1}{6}$	**O**	$\frac{7}{7} + \frac{8}{8} + \frac{9}{9}$

○	○	○	○	○	○	○	○	○	○	○	○	○	○
$\frac{5}{7}$	2	$\frac{2}{5}$	3	$\frac{7}{9}$	1	$1\frac{1}{4}$	$\frac{3}{5}$	$1\frac{1}{5}$	$\frac{1}{3}$	$1\frac{1}{3}$	$1\frac{1}{2}$	$\frac{4}{5}$	$1\frac{1}{9}$	$\frac{2}{3}$	$\frac{1}{2}$



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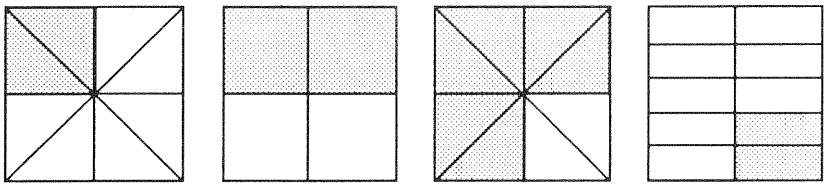
BREAK THE CODE



Each shaded fraction in its simplest form can be changed to a letter using the code box. The letters make up words and the words make up a question.
Find the answer to the question.

CODE	
A	$\frac{3}{4}$
B	1
C	$\frac{3}{8}$
D	$\frac{1}{6}$
E	$\frac{5}{8}$
F	$\frac{2}{3}$
G	2
H	$\frac{1}{2}$
I	$\frac{2}{5}$
J	$\frac{5}{6}$
K	3
L	$\frac{1}{8}$
M	$\frac{7}{8}$
N	$\frac{1}{7}$
O	$\frac{1}{3}$
P	$\frac{1}{11}$
Q	$\frac{1}{9}$
R	$\frac{3}{7}$
S	$\frac{3}{5}$
T	$\frac{1}{5}$
U	5
V	4
W	$\frac{1}{4}$
X	$\frac{4}{5}$
Y	$\frac{1}{10}$
Z	$\frac{3}{16}$

FIRST WORD



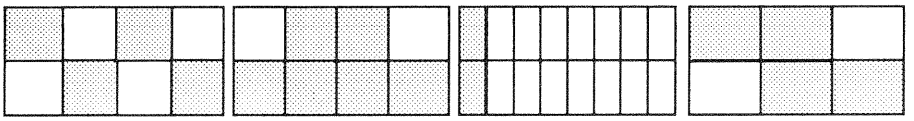
____ ____ ____ ____

SECOND WORD



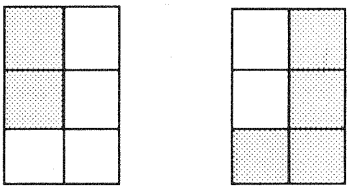
____ ____

THIRD WORD



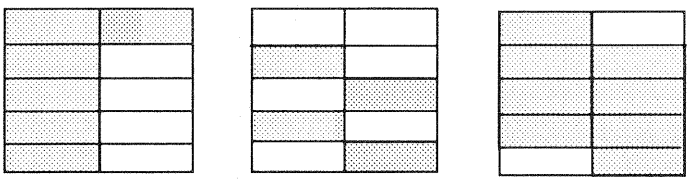
____ ____ ____ ____

FOURTH WORD



____ ____

FIFTH WORD



____ ____ ____

ANSWER: _____

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BREAK THE CODE

- *Each fraction in its simplest form can be changed to a letter using the code box.*
- *The letters make up words and the words make up a question.*
- *Find the answer to the question.*

1. First word $\frac{2}{6}$ $\frac{2}{16}$ $\frac{6}{16}$ $\frac{2}{4}$

Second word $\frac{2}{8}$ $\frac{6}{8}$

Third word $\frac{3}{6}$ $\frac{3}{9}$ $\frac{2}{10}$

Fourth word $\frac{4}{8}$ $\frac{4}{16}$ $\frac{10}{16}$ $\frac{4}{6}$ $\frac{12}{16}$

Fifth word $\frac{8}{16}$ $\frac{6}{9}$ $\frac{14}{16}$

ANSWER _ _ _ _ _ _ _ _ _ _

2. First word $\frac{2}{8}$ $\frac{9}{12}$

Second word $\frac{3}{15}$ $\frac{14}{16}$ $\frac{4}{6}$

Third word $\frac{2}{16}$ $\frac{9}{24}$ $\frac{6}{10}$ $\frac{2}{20}$

Fourth word $\frac{30}{32}$ $\frac{3}{12}$ $\frac{4}{10}$ $\frac{6}{15}$ $\frac{8}{12}$ $\frac{8}{10}$

Fifth word $\frac{2}{4}$ $\frac{3}{24}$ $\frac{12}{32}$ $\frac{21}{24}$

Sixth word $\frac{4}{20}$ $\frac{28}{32}$ $\frac{10}{15}$

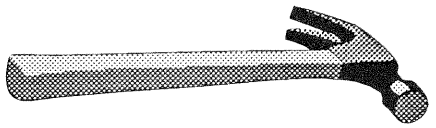
Seventh word $\frac{5}{10}$ $\frac{4}{32}$ $\frac{5}{20}$ $\frac{12}{15}$ $\frac{2}{32}$

ANSWER _ _ _ _ _ _ _ _ _

CODE	
A	$\frac{3}{8}$
B	$\frac{15}{16}$
C	2
D	$\frac{1}{16}$
E	$\frac{2}{3}$
F	$\frac{1}{10}$
G	$\frac{2}{5}$
H	$\frac{1}{8}$
I	$\frac{1}{4}$
J	$\frac{1}{7}$
K	$\frac{2}{7}$
L	$\frac{3}{5}$
M	$\frac{5}{8}$
N	$\frac{7}{8}$
O	$\frac{1}{5}$
P	7
Q	$\frac{1}{20}$
R	$\frac{4}{5}$
S	$\frac{3}{4}$
T	$\frac{1}{2}$
U	$1\frac{1}{4}$
V	$3\frac{1}{2}$
W	$\frac{1}{3}$
X	$\frac{5}{6}$
Y	$\frac{6}{7}$
Z	1

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BREAK THE CODE





The answer to each sum can be changed to a letter using the code box. The letters make up the words and the words make up a question.
FIND THE ANSWER TO THE QUESTION.

ANSWER 1 _ _ _ _ _ _ _ _ _ _ _ _ _ _ _ _

2 _ _ _ _ _ _ _ _ _ _ _ _ _ _ _ _

A	$\frac{3}{4}$
B	$\frac{7}{8}$
C	1
D	$\frac{5}{6}$
E	$\frac{1}{4}$
F	$\frac{3}{10}$
G	$\frac{4}{5}$
H	$\frac{2}{3}$
I	$\frac{1}{3}$
J	$1\frac{1}{2}$
K	$2\frac{2}{3}$
L	$\frac{7}{10}$
M	$\frac{3}{8}$
N	$\frac{1}{2}$
O	$\frac{2}{5}$
P	$\frac{5}{8}$
Q	$\frac{1}{8}$
R	$\frac{3}{7}$
S	$\frac{1}{5}$
T	$\frac{5}{7}$
U	$\frac{1}{7}$
V	$\frac{11}{12}$
W	$\frac{3}{5}$
X	$\frac{1}{32}$
Y	$\frac{1}{6}$
Z	$2\frac{1}{2}$

1.

FIRST WORD

$\frac{1}{3}+\frac{1}{3}$ $\frac{1}{5}+\frac{1}{5}$ $\frac{1}{5}+\frac{2}{5}$

SECOND WORD

$\frac{1}{16}+\frac{5}{16}$ $\frac{5}{8}+\frac{1}{8}$ $\frac{1}{4}+\frac{1}{4}$ $\frac{1}{12}+\frac{1}{12}$

THIRD WORD

$\frac{3}{8}+\frac{1}{8}$ $\frac{1}{6}+\frac{1}{6}$ $\frac{5}{16}+\frac{3}{16}$ $\frac{1}{8}+\frac{1}{8}$ $\frac{1}{10}+\frac{1}{10}$

FOURTH WORD

$\frac{3}{10}+\frac{1}{10}$ $\frac{7}{16}+\frac{1}{16}$

FIFTH WORD

$\frac{5}{14}+\frac{5}{14}$ $\frac{5}{9}+\frac{1}{9}$ $\frac{1}{9}+\frac{2}{9}$ $\frac{3}{20}+\frac{1}{20}$

SIXTH WORD

$\frac{5}{16}+\frac{5}{16}$ $\frac{11}{16}+\frac{1}{16}$ $\frac{7}{10}+\frac{1}{10}$ $\frac{1}{16}+\frac{3}{16}$

2.

FIRST WORD

$\frac{2}{3}-\frac{1}{3}$ $\frac{3}{5}-\frac{2}{5}$

SECOND WORD

$\frac{3}{5}-\frac{1}{5}$ $\frac{3}{4}-\frac{1}{4}$ $\frac{3}{8}-\frac{1}{8}$

THIRD WORD

$\frac{3}{16}-\frac{1}{16}$ $\frac{3}{14}-\frac{1}{14}$ $\frac{7}{8}-\frac{1}{8}$ $\frac{6}{7}-\frac{3}{7}$ $\frac{6}{7}-\frac{1}{7}$

$\frac{7}{8}-\frac{5}{8}$ $\frac{9}{14}-\frac{3}{14}$

FOURTH WORD

$\frac{15}{16}-\frac{1}{16}$ $\frac{8}{9}-\frac{5}{9}$ $\frac{9}{10}-\frac{1}{10}$ $\frac{14}{15}-\frac{2}{15}$ $\frac{5}{16}-\frac{1}{16}$

$\frac{11}{14}-\frac{5}{14}$

FIFTH WORD

$\frac{11}{14}-\frac{1}{14}$ $\frac{7}{9}-\frac{1}{9}$ $\frac{15}{16}-\frac{3}{16}$ $\frac{5}{8}-\frac{1}{8}$

SIXTH WORD

$\frac{7}{10}-\frac{3}{10}$ $\frac{9}{16}-\frac{1}{16}$ $\frac{5}{8}-\frac{3}{8}$

SEVENTH WORD

$\frac{13}{14}-\frac{3}{14}$ $\frac{5}{6}-\frac{1}{6}$ $\frac{11}{12}-\frac{7}{12}$ $\frac{20}{21}-\frac{11}{21}$ $\frac{11}{12}-\frac{1}{12}$

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BREAK THE CODE

1. FIRST WORD

$1 + \frac{3}{5}$ $1 + \frac{1}{3}$ $1 + \frac{1}{4}$ $1\frac{1}{2} + \frac{1}{2}$

SECOND WORD

$\frac{3}{5} + \frac{4}{5}$ $\frac{6}{11} + \frac{6}{11}$

THIRD WORD

$\frac{3}{7} + \frac{6}{7}$ $1 + \frac{1}{5}$ $1 - \frac{1}{4}$ $1 - \frac{2}{5}$

FOURTH WORD

$1\frac{1}{3} + \frac{2}{3}$ $\frac{4}{5} + \frac{3}{5}$ $\frac{3}{4} + \frac{3}{4}$ $\frac{1}{2} + \frac{1}{2}$ $\frac{7}{11} + \frac{5}{11}$

FIFTH WORD

$\frac{8}{11} + \frac{4}{11}$ $\frac{7}{10} + \frac{7}{10}$ $1 - \frac{1}{2}$

Find the answer to each sum in the code box below. The matching letters form the words of questions. What are the answers to these questions?

2. FIRST WORD

$\frac{2}{3} + \frac{2}{3}$ $\frac{3}{5} + \frac{3}{5}$ $\frac{4}{5} + \frac{4}{5}$

SECOND WORD

$\frac{7}{8} + \frac{5}{8}$ $\frac{7}{8} + \frac{3}{8}$ $\frac{5}{9} + \frac{5}{9}$ $1 - \frac{1}{3}$

THIRD WORD

$\frac{2}{5} + \frac{3}{5}$ $\frac{14}{15} + \frac{7}{15}$ $1 - \frac{3}{4}$ $\frac{7}{9} + \frac{5}{9}$ $2\frac{1}{2} - \frac{1}{2}$ $\frac{8}{9} + \frac{4}{9}$ $\frac{9}{11} + \frac{3}{11}$

FOURTH WORD

$\frac{4}{5} + \frac{2}{5}$ $\frac{4}{7} + \frac{5}{7}$

FIFTH WORD

$\frac{4}{7} + \frac{4}{7}$ $\frac{5}{8} + \frac{5}{8}$ $\frac{7}{8} + \frac{7}{8}$ $\frac{2}{3} + \frac{1}{3}$

SIXTH WORD

$\frac{11}{12} + \frac{7}{12}$ $\frac{3}{4} + \frac{1}{4} + \frac{1}{4}$ $\frac{1}{4} + \frac{3}{4} + \frac{3}{4}$ $\frac{3}{8} + \frac{5}{8}$

SEVENTH WORD

$\frac{9}{10} + \frac{3}{10}$ $\frac{8}{9} + \frac{2}{9}$ $\frac{1}{4} + \frac{3}{4}$

EIGHTH WORD

$\frac{9}{10} + \frac{7}{10}$ $\frac{11}{18} + \frac{13}{18}$ $\frac{4}{15} + \frac{14}{15}$ $1 - \frac{1}{5}$ $\frac{5}{16} + \frac{11}{16}$

NINTH WORD

$2 - \frac{6}{7}$ $2 - \frac{3}{4}$ $2 - \frac{1}{4}$ $\frac{7}{10} + \frac{3}{10}$



ANSWER 1 _____ **ANSWER 2** _____

A	B	C	D	E	F	G	H	I	J	K	L	M
$1\frac{1}{4}$	$\frac{1}{3}$	$1\frac{1}{7}$	$\frac{3}{10}$	1	$1\frac{2}{7}$	$\frac{1}{4}$	$1\frac{1}{3}$	$1\frac{2}{5}$	$1\frac{5}{6}$	$1\frac{3}{4}$	$\frac{4}{5}$	$1\frac{1}{2}$

CODE BOX

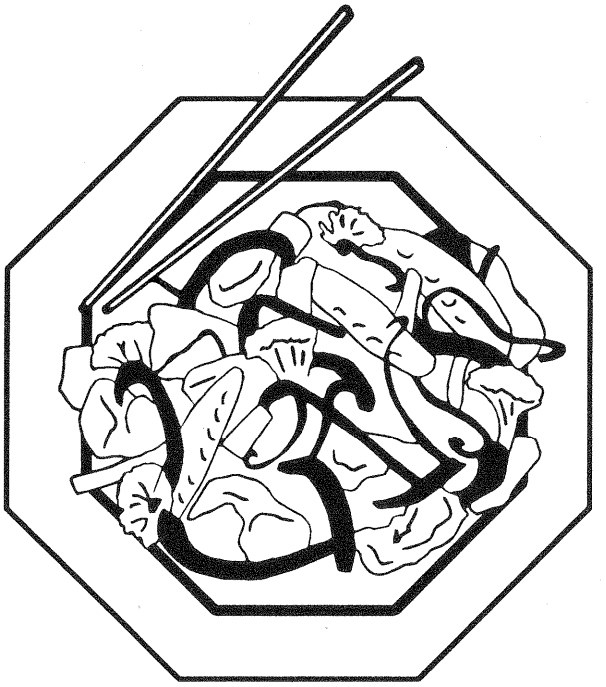
N	O	P	Q	R	S	T	U	V	W	X	Y	Z
$1\frac{1}{9}$	$1\frac{1}{5}$	7	$\frac{1}{16}$	$\frac{3}{5}$	$1\frac{1}{11}$	2	$\frac{3}{4}$	$\frac{5}{8}$	$1\frac{3}{5}$	$\frac{1}{2}$	$\frac{2}{3}$	8

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WHAT FLIES AND IS FULL OF NOODLES?

Determine each of the following. Look for your answers in the puzzle below. Write the letter of each question above its answer.





U $\frac{1}{2}$ of 12

M $\frac{1}{4}$ of 12

P $\frac{1}{3}$ of 15

O $\frac{1}{5}$ of 10

N $\frac{1}{4}$ of 36

R $\frac{1}{5}$ of 50

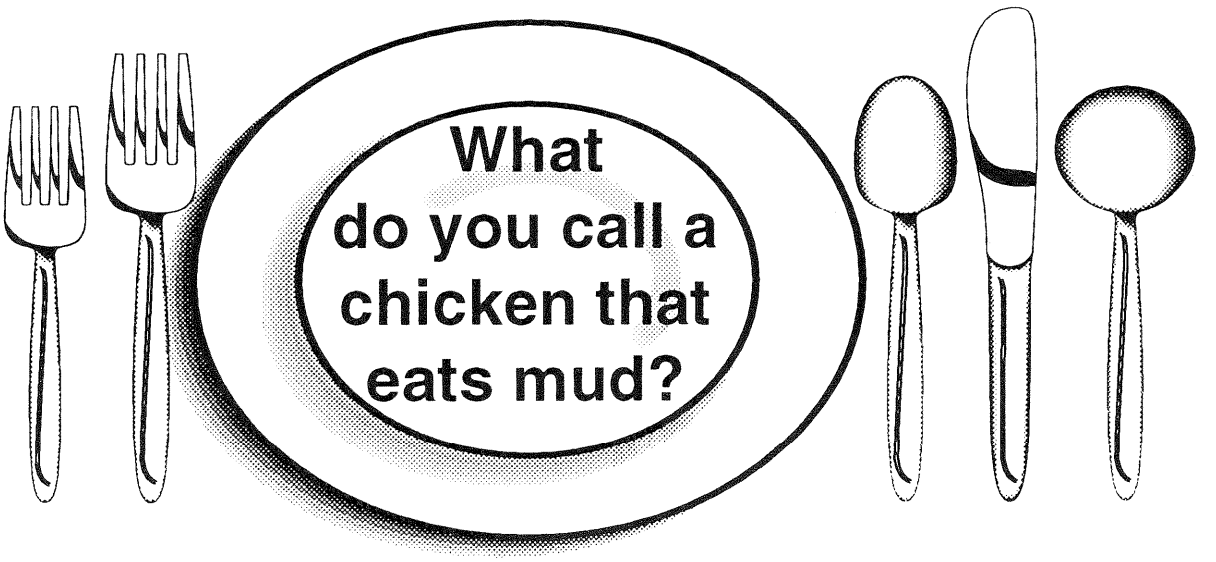
S $\frac{1}{4}$ of 28

A $\frac{1}{4}$ of 16

E $\frac{1}{3}$ of 24

○	○	○	○	○	○	○	○	○
7	2	6	5	8	10	3	4	9

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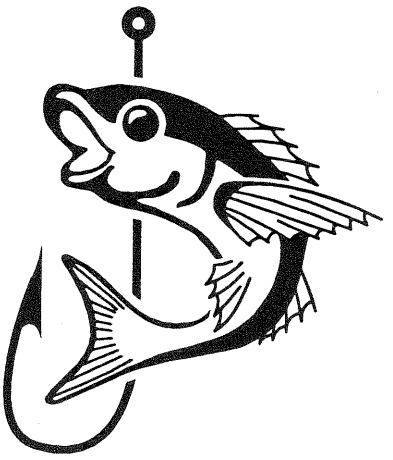
5 9 4 11 6 3 8 10 1 7 2

Solve each problem in the left hand column. Look for your answers in the right hand column. Rule a line to join each pair. Each line will pass through a number and a letter. Write the letter above the number in the puzzle code.

$\frac{1}{2}$ of 14	20
$\frac{1}{3}$ of 24	6
$\frac{2}{3}$ of 9	12
$\frac{3}{4}$ of 20	24
$\frac{3}{8}$ of 48	7
$\frac{3}{4}$ of 32	21
$\frac{2}{5}$ of 50	14
$\frac{3}{5}$ of 35	8
$\frac{5}{6}$ of 30	18
$\frac{2}{3}$ of 21	25
$\frac{3}{4}$ of 16	15

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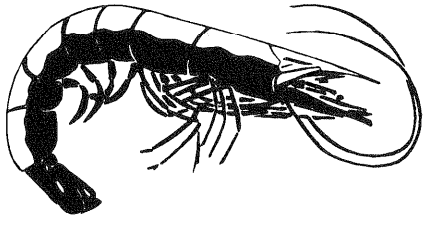
How Do You Get In Touch With A Fish?



Solve each problem in the left hand column. Look for your answers in the right hand column. Rule a line to join each pair. Each line will pass through a number and a letter. Write the letter above the number in the puzzle code.

$\frac{1}{2}$ of 10 ★	★ 20
$\frac{4}{5}$ of 20 ★	★ 6
$\frac{7}{8}$ of 24 ★	★ 5
$\frac{1}{3}$ of 12 ★	★ 18
$\frac{1}{4}$ of 8 ★	★ 9
$\frac{3}{4}$ of 8 ★	★ 16
$\frac{2}{5}$ of 20 ★	★ 12
$\frac{7}{8}$ of 40 ★	★ 30
$\frac{5}{6}$ of 30 ★	★ 4
$\frac{3}{4}$ of 12 ★	★ 25
$\frac{2}{3}$ of 27 ★	★ 2
$\frac{3}{5}$ of 50 ★	★ 35
$\frac{3}{4}$ of 32 ★	★ 21
$\frac{4}{5}$ of 25 ★	★ 24
$\frac{3}{4}$ of 16 ★	★ 8

5 T S 12 7 10 4 E L 14 I J U 6 P I T 13 1 N 9 3 15 A R 8 O 2 D



11 4 12 5 2 15 10 6

14 1 8 13 9 3 7

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WHY IS WAITING ON THE TELEPHONE LIKE DOING A TRAPEZE ACT?

To solve the puzzle, find the answer to each question and then look up the code box to find the letter that goes with the answer. Write the letter above the answer in the space provided below.

E. $\frac{1}{2}$ of 124

E. $\frac{1}{3}$ of 156

A. $\frac{1}{4}$ of 128

Y. $\frac{1}{5}$ of 400

O. $\frac{2}{3}$ of 99

T. $\frac{3}{4}$ of 108

C. $\frac{4}{5}$ of 85

O. $\frac{3}{8}$ of 96

B. $\frac{1}{9}$ of 45

H. $\frac{1}{7}$ of 56

U. $\frac{9}{10}$ of 400

A. $\frac{3}{5}$ of 150

N. $\frac{1}{6}$ of 300

E. $\frac{1}{8}$ of 96

O. $\frac{3}{10}$ of 180

S. $\frac{2}{5}$ of 65

G. $\frac{6}{7}$ of 49

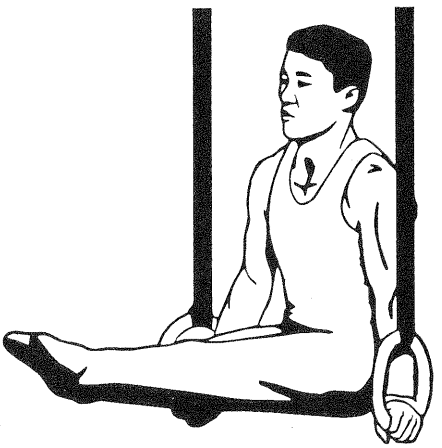
A. $\frac{7}{8}$ of 800

V. $\frac{3}{4}$ of 200

U. $\frac{2}{9}$ of 171

H. $\frac{5}{6}$ of 72

N. $\frac{3}{5}$ of 85

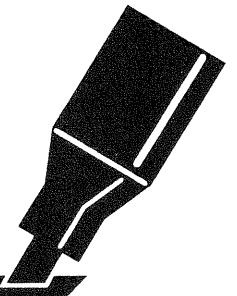




	5	12	68	32	38	26	52	80	54	360	
8	700	150	62	81	66	60	90	51	42	36	50

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Work out the answers to the following fraction problems.

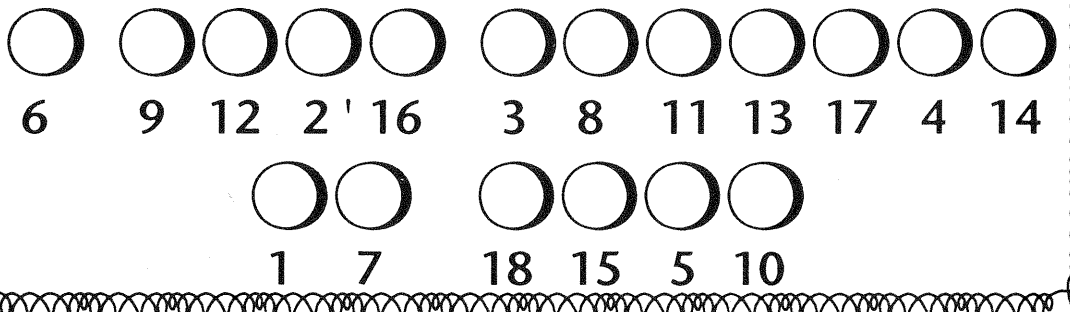


Match the question number to the letter that is next to the answer in order to find the solution to the riddle.

WHAT DID MR SPOCK SAY WHEN HE FIRST LOOKED IN THE MIRROR?

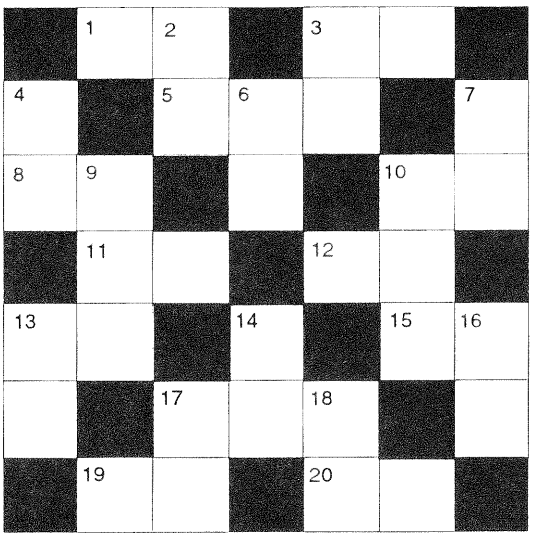
1. $\frac{1}{2}$ of 60
2. $\frac{1}{3}$ of 150
3. $\frac{1}{4}$ of 80
4. $\frac{1}{5}$ of 75
5. $\frac{1}{10}$ of 70
6. $\frac{1}{8}$ of 32
7. $\frac{3}{5}$ of 100
8. $\frac{3}{4}$ of 16
9. $\frac{2}{3}$ of 60
10. $\frac{5}{8}$ of 72
11. $\frac{7}{10}$ of 110
12. $\frac{4}{5}$ of 95
13. $\frac{5}{8}$ of 40
14. $\frac{1}{6}$ of 84
15. $\frac{5}{6}$ of 90
16. $\frac{1}{7}$ of 91
17. $\frac{3}{7}$ of 112
18. $\frac{5}{7}$ of 77

CODE	
B	20
I	4
E	12
S	45
A	76
E	14
E	48
M	30
C	40
L	77
I	25
T	13
Y	60
N	50
V	15
E	55
A	75
R	7



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'HALVES OF' CROSSNUMBER



REMEMBER
0·5 and 50% are just other names for one half.

ACROSS

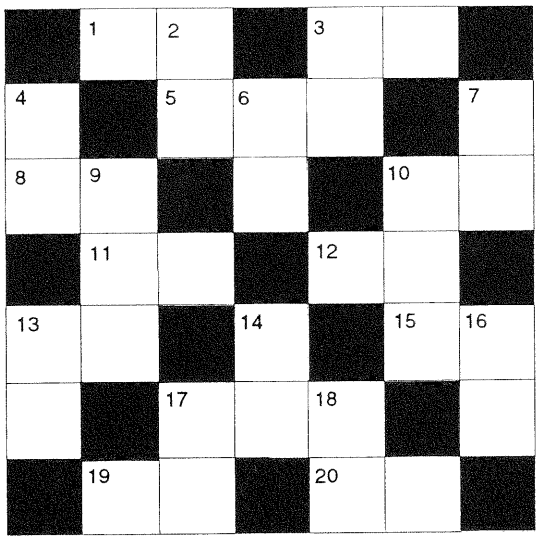
1. Half of 64
3. Half of a metre (in cm)
5. 1280 x $\frac{1}{2}$
8. Half of a half of 100
10. Half a dozen times 10
11. Halve 28 and add 4
12. Share $70 among two
13. Half of half of 160
15. Half of 66 doubled
17. A Jumbo Jet
19. 50% of 134
20. Half a million divided by 10 000

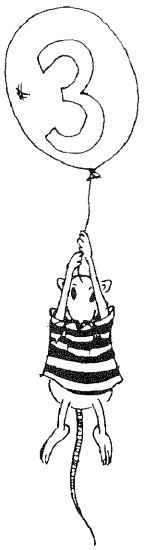
DOWN

2. The number of fortnights in a year
3. Half a dollar (in cents)
4. Half a day (in hours)
6. $\frac{1}{2}$ of 88
7. Half an hour (in mins)
9. $\frac{1}{2}$ x 1020
10. Double this number is 1312
13. 50% of 94
14. Half the number of days in February
16. 132 x 0·5
17. Double this number is 154
18. Half of half of 300

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'THIRDS OF' CROSSNUMBER





ACROSS

1. One third of 99

3. Share $186 evenly among 3 people

5. $\frac{1}{3}$ of 1635

8. one third of 282

10. $\frac{1}{3}$ x 177

11. one third of a minute (in seconds)

12. 210 x $\frac{1}{3}$

13. Share $105 evenly among 3 people

15. Three times this number is 189

17. One third of 1035

19. $\frac{2}{3}$ x 24

20. one third of the days of November

DOWN

2. $\frac{1}{3}$ of 105

3. one third of 195

4. 297 x $\frac{1}{3}$

6. one third of $\frac{1}{3}$ of 360

7. $\frac{1}{3}$ x 87

9. one third of 1275

10. one third of 1518

13. one third of $\frac{1}{3}$ of 270

14. one third of 132

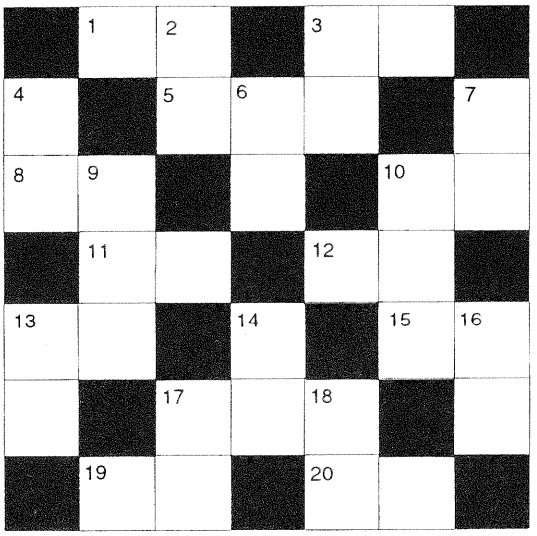
16. $\frac{1}{3}$ of 117

17. $\frac{2}{3}$ of 54

18. Three times this number is 153

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'QUARTERS OF' CROSSNUMBER



ACROSS

1. One quarter of 88
3. $\frac{1}{4}$ x 240
5. 2100 x $\frac{1}{4}$
8. Four times this number is sixty
10. $\frac{1}{4}$ of 128
11. 396 x one quarter
12. One quarter of 300
13. $\frac{1}{4}$ x 180
15. $\frac{1}{4}$ of 100
17. $\frac{1}{4}$ of 1768
19. Four times this number is 120
20. One quarter of $\frac{1}{4}$ of 640

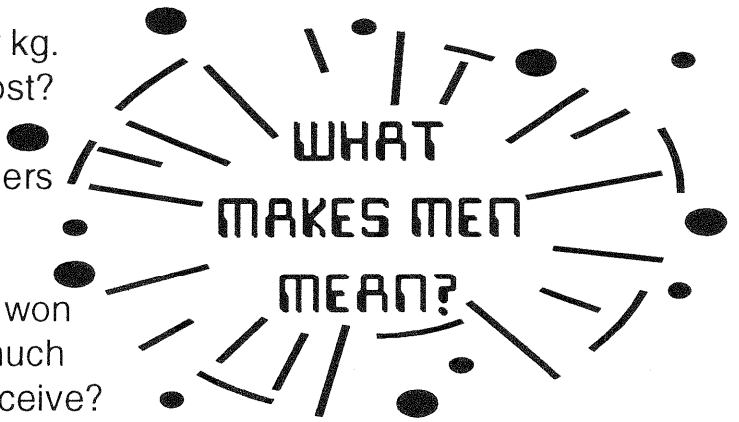
DOWN

2. $\frac{1}{4}$ of 100
3. One quarter of 260
4. Share $364 evenly among 4 people
6. $\frac{1}{4}$ of 112
7. $\frac{1}{4}$ of 248
9. 2380 ÷ 4
10. One quarter of 1408
13. $\frac{3}{4}$ of 56
14. Share $336 evenly among 4 people
16. Three quarters of 76
17. One quarter of 160
18. $\frac{3}{4}$ x 32

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WORK OUT THE ANSWERS TO THE FOLLOWING PROBLEMS. NOTE THE QUESTION NUMBER AND MATCH IT TO THE ANSWER IN THE CODE BOX. USE THIS INFORMATION TO ANSWER THE RIDDLE:

1. Bananas cost $1·50 per kg. How much will 2·4 kg cost?
2. Find the cost of 18 staplers if they cost $2·35 each.
3. A syndicate of 8 people won $276 in a lottery. How much should each member receive?



4. Twelve equal repayments are to be made on a car costing $4 320. How much is each repayment?
5. How much change should you receive from $100 if you buy a jumper worth $77·70?
6. Six items worth $14·10, $18·45, $63·60, $0·35, $6·70 and $27·80 were bought at a supermarket. What was the total bill?
7. Thirty items, each worth 65c were bought with a $50 note. How much change should you receive?
8. Donations of $2·00, 85c, $1·30, $2·10 and 90c were made to buy a gift worth $9·90. How much more needed to be collected?
9. Sue earns $562·50 for working 25 hours. How much does she earn per hour?
10. Hein receives $3·50 per week pocket money. If he saves half that amount, how much will he have saved in nine weeks?

CODE	
R	$34·50
T	$22·30
E	$131·00
E	$2·75
H	$42·30
T	$22·50
T	$360
L	$3·60
E	$30·50
A	$15·75

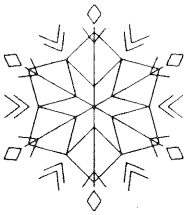
| | | |

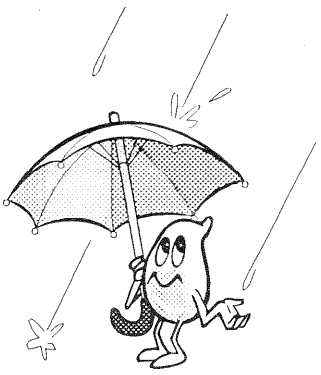
PREDICTING PATTERNS 1

DO THE FIRST THREE PROBLEMS IN EACH SET AND THEN TRY TO PREDICT THE REST. USE YOUR CALCULATOR TO CHECK YOUR PREDICTIONS. TRY TO DESCRIBE THE PATTERN THAT IS FORMED IN WORDS AND MATCH IT TO THE DESCRIPTION GIVEN ON THE NEXT PAGE. NOTE THE NUMBER NEXT TO THE QUESTION AND THE LETTER NEXT TO IT TO FIND THE ANSWER TO THE QUESTION:

What Do Fleas Do In Winter?

1.
37 037 x 3 = _ _ _ _
37 037 x 6 = _ _ _ _
37 037 x 9 = _ _ _ _
37 037 x 12 = _ _ _ _
37 037 x 15 = _ _ _ _
37 037 x 18 = _ _ _ _
37 037 x 21 = _ _ _ _
37 037 x 24 = _ _ _ _
37 037 x 27 = _ _ _ _





2.
2 222 222 x 9 = _ _ _ _
3 333 333 x 9 = _ _ _ _
4 444 444 x 9 = _ _ _ _
5 555 555 x 9 = _ _ _ _
6 666 666 x 9 = _ _ _ _
7 777 777 x 9 = _ _ _ _
8 888 888 x 9 = _ _ _ _
9 999 999 x 9 = _ _ _ _

3.
4 x 2 178 = _ _ _ _ _ _ _ _ _
4 x 21 978 = _ _ _ _ _ _ _ _ _
4 x 219 978 = _ _ _ _ _ _ _ _ _
4 x 2 199 978 = _ _ _ _ _ _ _ _
4 x 21 999 978 = _ _ _ _ _ _ _

4.
9 x 1 089 = _ _ _ _ _ _ _ _ _ _
9 x 10 989 = _ _ _ _ _ _ _ _ _ _
9 x 109 989 = _ _ _ _ _ _ _ _ _
9 x 1 099 989 = _ _ _ _ _ _ _ _
9 x 10 999 989 = _ _ _ _ _ _ _

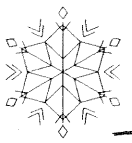


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PREDICTING PATTERNS 1 (Contd.)

5.
99 x 12 = _ _ _ _ _ _
99 x 23 = _ _ _ _ _ _
99 x 34 = _ _ _ _ _ _ _
99 x 45 = _ _ _ _ _ _ _
99 x 56 = _ _ _ _ _ _ _
99 x 67 = _ _ _ _ _ _ _
99 x 78 = _ _ _ _ _ _ _
99 x 89 = _ _ _ _ _ _ _

6.
9109 x 1 = _ _ _ _ _
9109 x 2 = _ _ _ _ _
9109 x 3 = _ _ _ _ _
9109 x 4 = _ _ _ _ _
9109 x 5 = _ _ _ _ _
9109 x 6 = _ _ _ _ _
9109 x 7 = _ _ _ _ _
9109 x 8 = _ _ _ _ _
9109 x 9 = _ _ _ _ _



PATTERN DESCRIPTIONS

E The pattern consists of multiples of nine at either end and the number you multiplied by in the middle.

F The pattern is made up entirely of the same digit depending on which multiple you are using.

E The first two numbers in this pattern are always 98 and the last number is 1. The number of 9's in the middle of the number depend on the multiple with which you are dealing.

E The first two numbers in this pattern are always 87 and the last two are always 12. The number of 9's in the middle of the number depend on the multiple with which you are dealing.

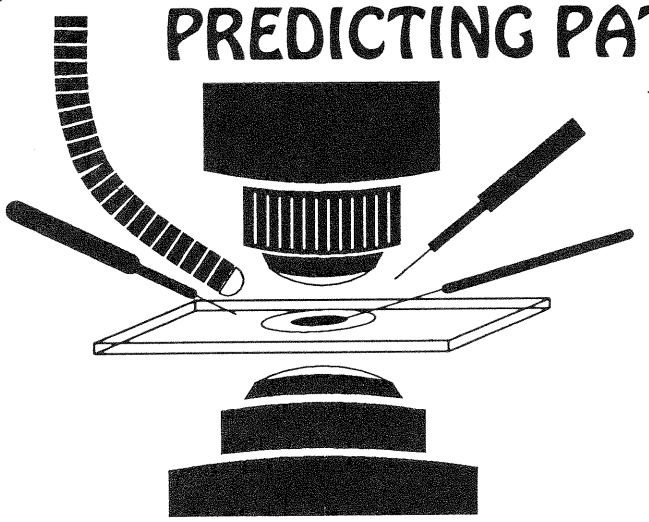
L This pattern is made up almost entirely of nines. Even the first and last digit of the pattern produced add up to make 9.

Z The first two digits and the last two digits in this pattern add up to 99.

1	2	4	6	5	3

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PREDICTING PATTERNS 2

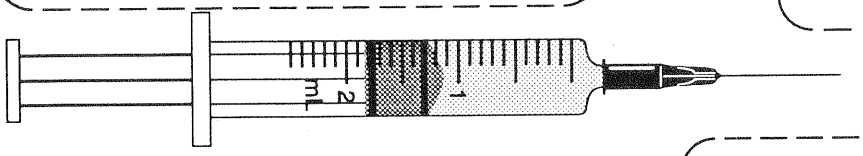


Try the first three problems in each set and then try to predict the rest. Use your calculator to check your predictions. Try to describe the pattern that is formed in words and match it to the description given on the following page. Note the number next to the question and the letter next to the description to find the answer to :

What is the most dangerous part of a needle?

1. 3367 x 33 = _____
   3367 x 66 = _____
   3367 x 99 = _____
   3367 x 132 = _____
   3367 x 165 = _____
   3367 x 198 = _____
   3367 x 231 = _____
   3367 x 264 = _____
   3367 x 297 = _____

3. $\frac{1}{99}$ = _____
   $\frac{2}{99}$ = _____
   $\frac{3}{99}$ = _____
   $\frac{4}{99}$ = _____
   $\frac{5}{99}$ = _____
   $\frac{6}{99}$ = _____
   $\frac{7}{99}$ = _____
   $\frac{8}{99}$ = _____
   $\frac{9}{99}$ = _____



2. 999 x 12 = _____
   999 x 23 = _____
   999 x 34 = _____
   999 x 45 = _____
   999 x 56 = _____
   999 x 67 = _____
   999 x 78 = _____
   999 x 89 = _____

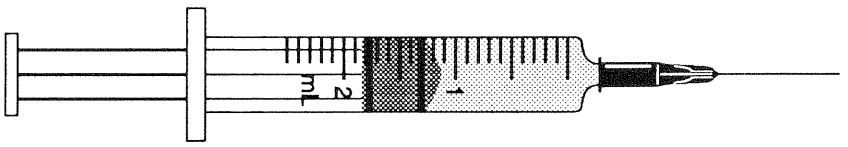
4. 99 x 13 = _____
   99 x 24 = _____
   99 x 35 = _____
   99 x 46 = _____
   99 x 57 = _____
   99 x 68 = _____
   99 x 79 = _____

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PREDICTING PATTERNS 2 (Contd.)

5. 1 x 1 = _____
11 x 11 = _____
111 x 111 = _____
1111 x 1111 = _____

6. 265 x 9 = _____
365 x 9 = _____
465 x 9 = _____
565 x 9 = _____
665 x 9 = _____
765 x 9 = _____
865 x 9 = _____



PATTERN DESCRIPTIONS

E A repeating pattern is formed where a single digit is separated by zeros.

H The first two digits and the last two digits add up to 99.

T The pattern is made up entirely of the same digit depending on which multiple you are using.

N The first two digits and the last two digits add up to 99 and are always separated by a 9.

E The last two digits of this pattern are always 8 and 5.

D The pattern starts by counting forwards in ones and finishes by counting backwards in ones.

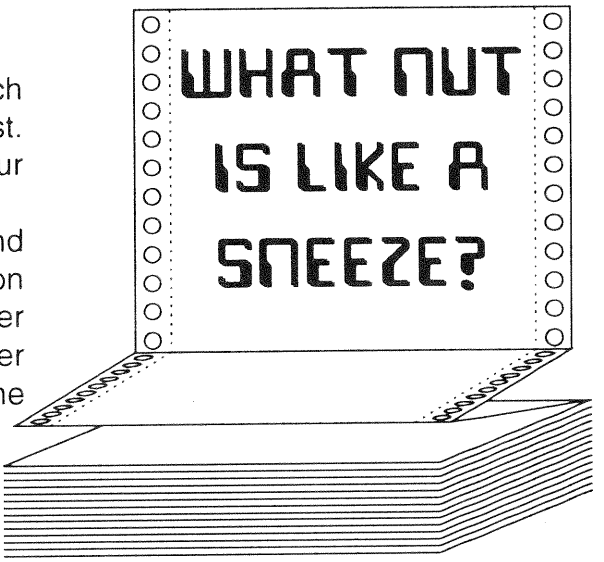
1	4	3		6	2	5



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Predicting Patterns 3

Try the first three problems in each set and then try to predict the rest. Use your calculator to check your predictions. Try to describe the pattern that is formed in words and match it to the description given on the following page. Note the number next to the question and the letter next to the description to find the answer to :



1.

15 873 x 7 = _ _ _ _ _
15 873 x 14 = _ _ _ _ _
15 873 x 21 = _ _ _ _ _
15 873 x 28 = _ _ _ _ _
15 873 x 35 = _ _ _ _ _
15 873 x 42 = _ _ _ _ _
15 873 x 49 = _ _ _ _ _
15 873 x 56 = _ _ _ _ _
15 873 x 63 = _ _ _ _ _

2.

143 x 7 = _ _ _ _ _
143 x 14 = _ _ _ _ _
143 x 21 = _ _ _ _ _
143 x 28 = _ _ _ _ _
143 x 35 = _ _ _ _ _
143 x 42 = _ _ _ _ _
143 x 49 = _ _ _ _ _
143 x 56 = _ _ _ _ _
143 x 63 = _ _ _ _ _

3.

199 x 11 = _ _ _ _ _
299 x 11 = _ _ _ _ _
399 x 11 = _ _ _ _ _
499 x 11 = _ _ _ _ _
599 x 11 = _ _ _ _ _
699 x 11 = _ _ _ _ _
799 x 11 = _ _ _ _ _
899 x 11 = _ _ _ _ _
999 x 11 = _ _ _ _ _

4.

6 x 7 = _ _ _ _ _ _ _
66 x 67 = _ _ _ _ _ _ _
666 x 667 = _ _ _ _ _ _
6666 x 6667 = _ _ _ _ _ _

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Predicting Patterns 3 (Contd.)

5.

91 x 11 = _ _ _ _ _ _

91 x 111 = _ _ _ _ _ _

91 x 1111 = _ _ _ _ _ _

91 x 11111 = _ _ _ _ _

91 x 111111 = _ _ _ _ _

6.

9 x 1 – 1 = _ _ _ _ _ _ _ _

9 x 21 – 1 = _ _ _ _ _ _ _ _

9 x 321 – 1 = _ _ _ _ _ _ _ _

9 x 4321 – 1 = _ _ _ _ _ _ _

9 x 54321 – 1 = _ _ _ _ _ _ _

9 x 654321 – 1 = _ _ _ _ _ _

9 x 7654321 – 1 = _ _ _ _ _ _

PATTERN DESCRIPTIONS

A The pattern always ends with 89.

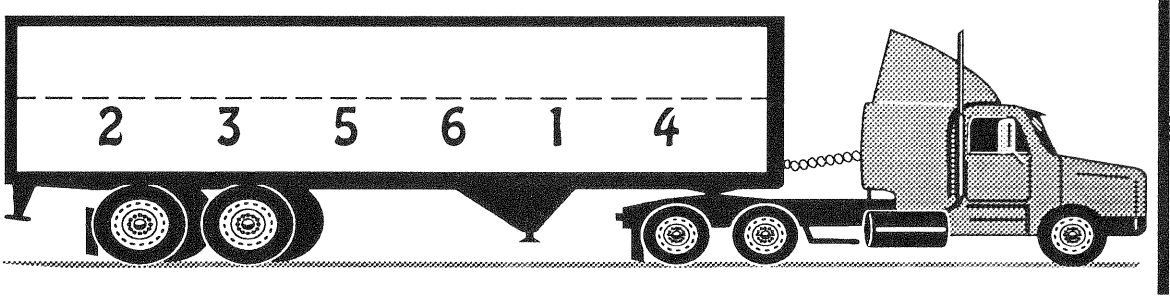
H The pattern consists almost entirely of eights.

S This pattern always begins and ends with one and always contains two zeros

E The pattern is made up entirely of the same digit depending on which multiple you are using.

W The pattern is always made up of fours and then twos. The number of fours and twos is always equal to the number of digits contained in the multipliers.

C The pattern begins and ends with the same digit and is separated by two zeros.



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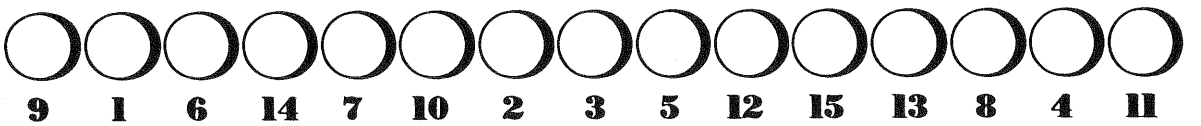
WHY DO BAKERS MAKE BREAD?

To solve the puzzle, find the next three numbers in each pattern and then look up the code box to find the letter that goes with the answer. Write the letter above the answer in the space provided.

1. 8, 11, 14, ____, ____, ____
2. 3, 6, 12, ____, ____, ____
3. 1, 4, 9, ____, ____, ____
4. 1, 1, 2, 3, 5, ____, ____, ____
5. 1, 2, 4, 8, ____, ____, ____
6. 1, 3, 9, 27, ____, ____, ____
7. 1, 8, 27, ____, ____, ____
8. 10, 100, 1000, ____, ____, ____
9. 1000, 100, 10, ____, ____, ____
10. 64, 32, 16, ____, ____, ____
11. 99, 92, 85, ____, ____, ____
12. 5, 11, 23, ____, ____, –
13. 4, 7, 13, ____, ____, ____
14. 1, 5, 25, ____, ____, ____
15. 6, 10, 14, ____, ____, ____

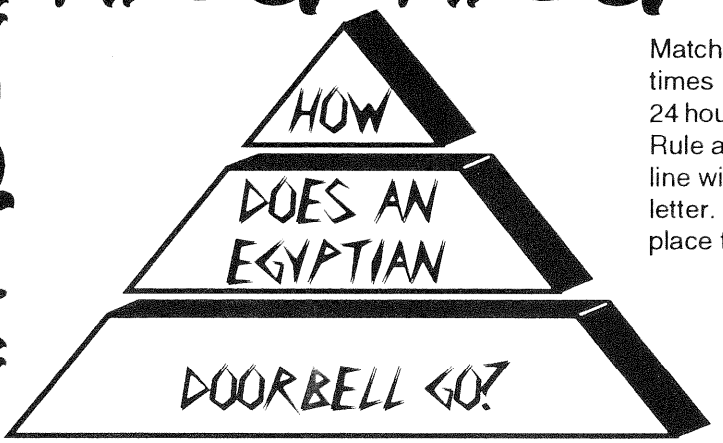
CODE BOX

D	18, 22,26
E	47, 95, 191
F	64, 125, 216
G	8, 13, 21
H	78, 71, 64
H	16, 32, 64
J	1, 0·1, 0·01
O	8, 4, 2
O	25, 49, 97
R	24, 48, 96
S	81, 243, 729
T	16, 25, 36
T	125, 625,3125
U	17, 20, 23
U	10 000, 100 000, 1 000 000



9 1 6 14 7 10 2 3 5 12 15 13 8 4 11

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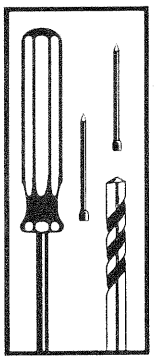
Match each of the following 12 hour times in the left hand column with the 24 hour times in the right hand column. Rule a line to connect each pair. Each line will pass through a number and a letter. The number tells you where to place the letter in the code below.

9.30 a.m.	0150
2.45 p.m.	1230
6.20 p.m.	1820
5.25 a.m.	0845
1.50 a.m.	1530
12.55 a.m.	0930
3.30 p.m.	1910
8.05 p.m.	0055
12.30 p.m.	2315
7.10 p.m.	0525
8.45 a.m.	0225
2.25 a.m.	2005
11.15 p.m.	1445

O 13 6 C 11 O O N N 7 3 A 1 D 10 T T 12 4 I 9 E 8 5 2 M

6 12 9 4 5 7 1 8 13 2 11 10 3

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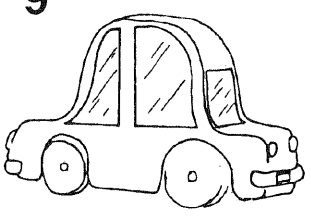


WHY DID THE MECHANIC SLEEP UNDER HIS CAR?

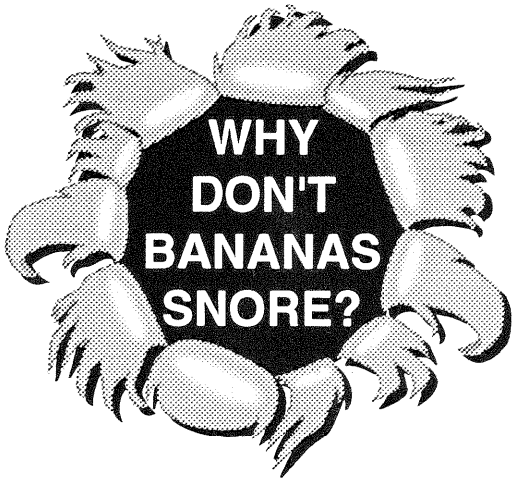
Match each of the following 24 hour times in the left hand column with the 12 hour times in the right hand column. Rule a line to join each pair. Each line will pass through a number and a letter. The number tells you where to place the letter in the code below.

2033 hrs	5.19 a.m.
1228 hrs	2.42 a.m.
0448 hrs	1.39 p.m.
2117 hrs	8.33 p.m.
2057 hrs	10.35 p.m.
0015 hrs	9.17 p.m.
0242 hrs	11.22 a.m.
1909 hrs	12.28 p.m.
2245 hrs	10.45 p.m.
0637 hrs	12.15 a.m.
1339 hrs	4.48 a.m.
1122 hrs	3.16 p.m.
0519 hrs	7.09 p.m.
1758 hrs	8.57 p.m.
1516 hrs	5.58 p.m.
2235 hrs	6.37 a.m.

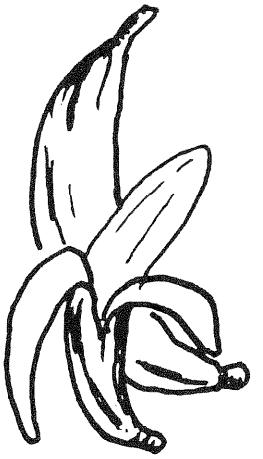
10	6	2	7	4	9	6	11	9	14	15	6	9	
12	13	14	5	16	3	5	4	9	10	6			
1	14	8	4	5	4	15							



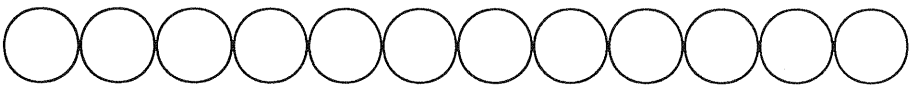
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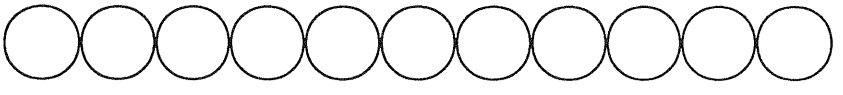
Complete each of the following conversions involving units of time. Look for your answers in the code below. Write the letter of each question above its answer.



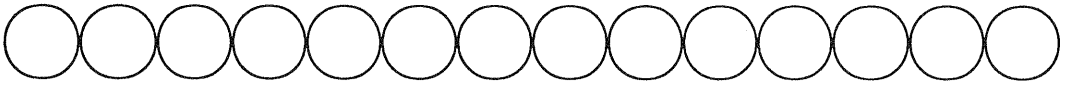
F 2 weeks = _ _ _ _ _ days
S 3 years = _ _ _ _ _ months
C $\frac{1}{2}$ year = _ _ _ _ _ months
Y 2 decades = _ _ _ _ years
U 7 weeks = _ _ _ _ _ days
B $\frac{1}{4}$ year = _ _ _ _ _ months
O 3 centuries = _ _ _ _ years
H Number of days in October, November and December _ _ _ _ _
R 5 fortnights = _ _ _ weeks
A 3 fortnights = _ _ _ days
E 12 weeks = _ _ _ _ days
T 4 years = _ _ _ _ months
N $1\frac{1}{2}$ years = _ _ _ _ months
W 7 decades = _ _ _ _ years
D 2 fortnights = _ _ _ _ days
P Number of days in any 4 consecutive years _ _ _ _ _ _
K 5 weeks = _ _ _ _ _ days



48 92 84 20 28 300 18 48 70 42 18 48

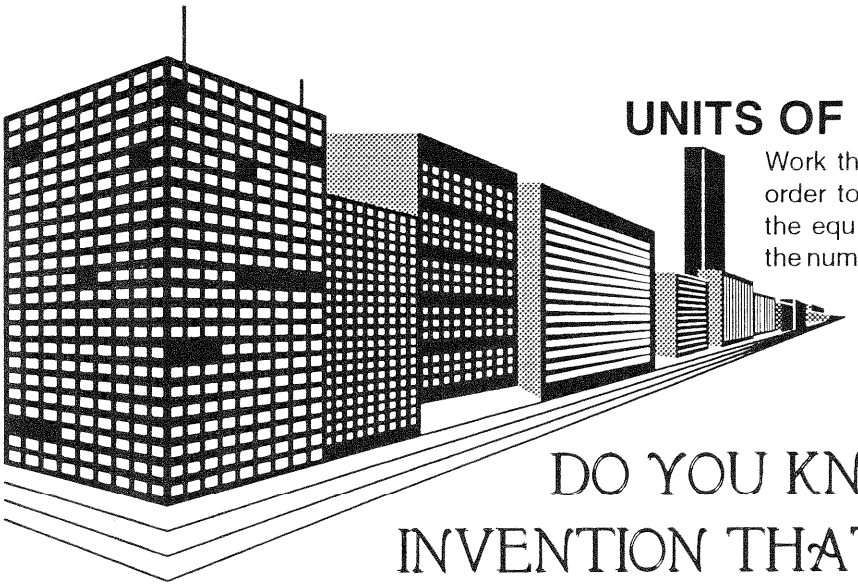


48 300 70 42 35 84 49 1461 48 92 84



10 84 36 48 300 14 48 92 84 3 49 18 6 92

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UNITS OF TIME

Work through the problems below in order to answer the question. Match the equivalent units of time and note the number and letter that go together.

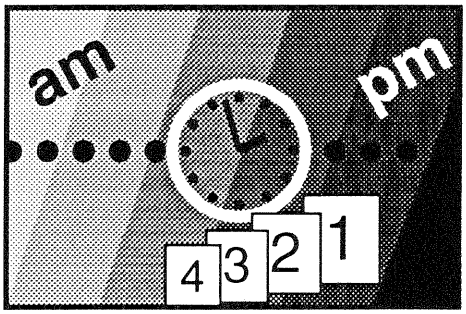
DO YOU KNOW OF ANY INVENTION THAT ALLOWS PEOPLE TO SEE THROUGH WALLS?

1.	24 hours
2.	1 hour
3.	30 seconds
4.	365 days
5.	$\frac{1}{2}$ hour
6.	1 fortnight
7.	$\frac{3}{4}$ hour
8.	120 seconds
9.	1 decade
10.	100 years
11.	3 days
12.	90 minutes
13.	15 minutes
14.	$\frac{1}{3}$ of an hour
15.	$\frac{1}{2}$ year

D	2 weeks
T	10 years
H	45 minutes
S	30 minutes
W	$\frac{1}{2}$ a minute
B	1 century
O	6 months
O	$1\frac{1}{2}$ hours
N	$\frac{1}{4}$ hour
I	20 minutes
O	1 day
U	1 year
A	72 hours
W	2 minutes
W	60 minutes

7	12	3	11	10	1	4	9	2	14	13	6	15	8	5	?

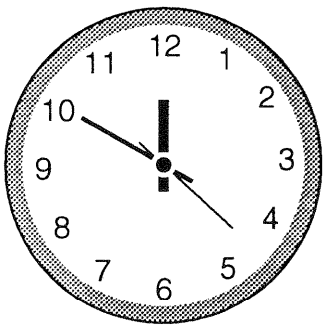
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WHEN DO CLOCKS BREAK DOWN?

15	210	6	300	72	180	600	360	90	150	36	96	30	45	120	48	20

Complete each of the following conversions involving units of time. Look for your answers in the code above. Write the letter of each question above its answer



H 3 minutes = ____ seconds

P $\frac{1}{3}$ minute = ____ seconds

E $\frac{1}{2}$ minute = ____ seconds

S 2 hours = ____ minutes

I $1\frac{1}{2}$ days = ____ hours

I 6 minutes = ____ seconds

W $\frac{1}{4}$ hour = ____ minutes

U 2 days = ____ hours

H $3\frac{1}{2}$ hours = ____ minutes

M 4 days = ____ hours

I $\frac{3}{4}$ hour = ____ minutes

E 10 minutes = ____ seconds

R $1\frac{1}{2}$ hours = ____ minutes

E $\frac{1}{4}$ day = ____ hours

N 5 hours = ____ minutes

T $2\frac{1}{2}$ minutes = ____ seconds

T 3 days = ____ hours

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DETERMINE THE NUMBER OF MINUTES THAT ELAPSE BETWEEN THE FOLLOWING PAIRS OF TIMES. LOOK FOR YOUR ANSWERS IN THE CODE KEY. NOTICE THE LETTER NEXT TO EACH ANSWER. WRITE THAT LETTER ABOVE THE NUMBER OF EACH QUESTION IN THE PUZZLE.

1. 10.15 a.m. to 10.47 a.m.
2. 3.06 p.m. to 3.51 p.m.
3. 10.55 p.m. to 11.18 p.m.
4. 4.45 a.m. to 5.28 a.m.
5. 11.58 p.m. to 12. 10 a.m.
6. 6.35 p.m. to 7.21 p.m.
7. 8.48 p.m. to 9.42 p.m.
8. 2.22 a.m. to 3.19 a.m.
9. 7.29 p.m. to 7.58 p.m.
10. 9.18 a.m. to 10.02 a.m.

CODE KEY

12 – *U*
23 – *N*
29 – *R*
32 – *H*
43 – *F*
44 – *L*
45 – *W*
46 – *B*
54 – *T*
57 – *E*

2	1	8	9	8	7	1	8	4	5	3	3	8	10	6	8

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MATH O MAGIC 1·2 SOLUTIONS

1. I'VE DONE A FINE DAY'S WORK.

2.

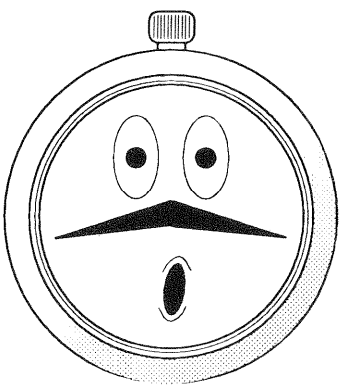
	1 9	2 ·1		3 3	9	
4 8		5 7	6 ·2	6		7 8
8 8	9 ·4		5		10 5	1
	11 9	·6		12 4	·8	
13 7	4		14 9		15 5	16 5
·6		17 1	·2	18 1		·6
	19 1	·6		20 ·8	8	

3.

	1 4	2 ·4		3 1	·1	
4 8		5 5	6 1	·2		7 2
8 6	9 2		·5		10 6	2
	11 ·1	5		12 2	4	
13 ·2	5		14 3		15 ·1	16 3
2		17 2	·4	18 5		·4
	19 4	·5		20 ·5	4	

4.

	1 9	2 ·9		3 ·4	9	
4 9		5 5	6 ·2	5		7 1
8 6	9 ·9		4		10 2	·5
	11 1	·9		12 9	·9	
13 1	1		14 3		15 5	16 ·4
·1		17 8	·2	18 5		5
	19 1	·1		20 6	·9	



5. THEY ALWAYS FEEL VINE

6. HOOPING COUGH

7. I'LL GRANT YOU THREE WASHES

8. O H 2 E C (OH TOO EASY

9. HE GOT A NASTY SHOCK

10. A FROG WITH HICCUPS

11. 3

12. 1. 20 2. YES

13. 1. 11 2. NO

14. 1. 24 2. 8

15. SOUPERMAN

16. A BRICKLAYER

17. JUST DROP IT A LINE

18. BECAUSE YOU HAVE TO HANG ON

MATH O MAGIC 1·2 SOLUTIONS (contd.)

19. I CAN'T BELIEVE MY EARS

20.

■	1 3	2 2	■	3 5	0	■
4 1	■	5 6	6 4	0	■	7 3
8 2	9 5	■	4	■	10 6	0
■	11 1	8	■	12 3	5	■
13 4	0	■	14 1	■	15 6	16 6
7	■	17 7	4	18 7	■	6
■	19 6	7	■	20 5	0	■

21.

■	1 3	2 3	■	3 6	2	■
4 9	■	5 5	6 4	5	■	7 2
8 9	9 4	■	0	■	10 5	9
■	11 2	0	■	12 7	0	■
13 3	5	■	14 4	■	15 6	16 3
0	■	17 3	4	18 5	■	9
■	19 1	6	■	20 1	0	■

22.

■	1 2	2 2	■	3 6	0	■
4 9	■	5 5	6 2	5	■	7 6
8 1	9 5	■	8	■	10 3	2
■	11 9	9	■	12 7	5	■
13 4	5	■	14 8	■	15 2	16 5
2	■	17 1	4	18 2	■	7
■	19 3	0	■	20 4	0	■



23. THE LETTER A
24. FLEEZE
25. THE END
26. CASHEW
27. JUST FOR THE DOUGH
28. TOOT AND COME IN
29. HE WANTED TO GET UP OILY IN THE MORNING
30. THEY DON'T WANT TO WAKE UP THE REST OF THE BUNCH
31. HOW ABOUT WINDOWS?
32. WHEN THEIR TIME IS UP
33. WHERE THE FUNNEL BE